Peer Evaluation Report

Presenter: Kylee Hoge **Course:** CSCI 443 – Introduction to Machine Learning **Project Title:** Predicting State Tax Revenues & Detecting Anomalies Using Machine Learning

Project Context & Motivation Kylee's presentation on predicting North Dakota tax revenues was really well-grounded. You could tell her actual experience working at the ND Tax Department gave her a solid grasp of the problem. She made a strong case for why this project matters, specifically explaining how the state's biannual legislative budget cycle makes accurate long-term forecasting critical for funding.

Data Processing & Transparency I thought her handling of the data was one of the strongest parts of the project. She clearly understood the "personality" of the data, such as how filing deadlines drive seasonality. What stood out to me was her transparency; she admitted that her first model failed to pick up on the seasonality of income tax. Instead of glossing over it, she explained how she fixed the issue by implementing One-Hot Encoding, which helped the model track those peaks much better.

Modeling Approaches Her comparison of the models was logical. For forecasting, finding that Linear Regression performed better than Random Forest made sense, given that the data was aggregated and linear. It seemed like the Random Forest model was overfitting the noise rather than catching the trend.

For anomaly detection, she gave a great explanation of why Z-Scores failed (because the data wasn't normally distributed) and why the Isolation Forest succeeded. The fact that the Isolation Forest was able to specifically catch COVID-19 disruptions in the hospitality sector was a really cool proof of concept.

Results & Delivery The presentation itself was clean and professional, and the graphs showing future forecasts against historical data were easy to read. Her delivery was calm and she did a good job translating technical stats into plain English.

regarding the results, she was very honest about the error rates. While the Sales Tax model was accurate (around 4% MAPE), the Income Tax model had a high error rate of around 80%. I actually liked that she didn't try to hide this; contextualizing the error relative to the billions collected added credibility to her work.

Project Pros & Cons

Pros

- **Real-World Application:** The project wasn't just theoretical; it was backed by genuine experience at the ND Tax Department, making the problem statement very convincing.

- **Adaptability:** I really appreciated that when the initial model failed to catch the seasonality of income tax, she didn't just leave it—she implemented One-Hot Encoding to fix it.
- **Strong Anomaly Detection:** The use of the Isolation Forest model was a win. It successfully identified specific real-world outliers, like the COVID-19 disruptions in the hospitality sector.
- **Honest Evaluation:** Instead of trying to hide the high error rates in the income tax model, she contextualized them well and explained the "why" behind the numbers, which added a lot of credibility.
- **Clear Communication:** The delivery was articulate and the slides were clean, specifically the "Future Forecast" graphs which clearly visualized the trends.

Cons

- **High Error in Income Tax:** While the sales tax model was solid, the income tax forecasting had a very high error rate (around 80% MAPE), showing that the current approach struggles with that specific dataset.
- **Model Limitations:** The Random Forest model didn't perform well for forecasting; it seemed to overfit the noise rather than capturing the actual trends.
- **Z-Score Failure:** The attempt to use Z-Scores for anomaly detection didn't work because the data didn't follow a normal distribution, though she did explain this limitation well.
- **Missing External Factors:** The models currently rely only on historical tax data. As noted in the critique, excluding external economic indicators (like oil prices or inflation) likely contributed to the lower accuracy in the income tax forecasts.

Suggestions for Future Work

- **Time-Series Specific Models:** Since the data is heavily seasonal, moving away from standard regression and trying models built for this like SARIMA or LSTMs might handle the sequential dependencies better.
- **External Factors:** As she noted in her "Future Work," the current models only look at historical tax data. Incorporating external economic indicators, like oil prices or inflation, would likely help bring down that high error rate in the Income Tax predictions.

Overall Impression This was a high-quality presentation that successfully bridged the gap between the machine learning concepts we learn in class and a real-world government application. The flow from data exploration to model correction made a complex topic very easy to follow.